SANTHOSH KUMAR M

.NET FULL STACK DEVELOPER | ENTERPRISE APPLICATIONS | MANUFACTURING DIGITALIZATION

Coimbatore, Tamil Nadu, India | +91 82704 20327 | mr.santhoshkumar4329@gmail.com
GitHub: github.com/santhoshkumar032001 | Portfolio: santhoshkumar032001.github.io/SamplePortfolio

PROFESSIONAL SUMMARY

.NET Full Stack Developer with 3+ years of experience designing, developing, deploying and maintaining business web applications using **C#, .NET 8, ASP.NET Core Web API, React.js, JavaScript and Microsoft SQL Server**. Experienced in manufacturing digitalization, shop-floor process tracking, QR-based traceability, IoT–FMS integration, stage-wise product image documentation, production dashboards and AI-assisted business process automation. Skilled in requirement analysis, API integration, SQL development, debugging, deployment and production support, with a track record of replacing manual operations with reliable digital workflows.

TECHNICAL SKILLS

Languages	C#, JavaScript (ES6+), SQL, HTML5, CSS3
Backend	.NET 8, ASP.NET Core, ASP.NET Web API, RESTful APIs
Frontend	React.js, Material UI (MUI), Axios, Responsive UI Development
Database	Microsoft SQL Server, SQL Query Development, Dapper
Reporting	Power BI Integration, Manufacturing Dashboards, Man-Hours-per-Ton Analysis, Booth-Wise Reporting
Documents & Images	PDF Handling, PDF.js, File Upload Integration, Stage-Wise Photo Documentation
Manufacturing Systems	FMS, QR Traceability, Machine Shop IoT, MPI Workflows, DP Requirements
Tools & Support	Visual Studio 2022, Git, GitHub, Browser DevTools, Web App Publishing, API Troubleshooting

PROFESSIONAL EXPERIENCE

Junior Software Engineer — March 2024 – Present

Peekay Steel Castings Pvt Ltd, Coimbatore

Full-Stack Application Development

- Design, develop and maintain enterprise web applications using React.js, Material UI, ASP.NET Core Web API, C# and SQL Server.
- Build responsive interfaces for manufacturing operations, production monitoring, process entry and reporting.
- Analyze requirements, implement enhancements and troubleshoot frontend, backend, API and SQL issues.

FMS Process Entry & Manufacturing Digitalization

- Developed and enhanced the FMS Process Entry Web Application, digitizing shop-floor data capture and process tracking.
- Replaced manual data capture with web-based workflows integrated with backend services and SQL Server.

QR Scanner & Production Traceability

- Built QR-based shop-floor scanning workflows with process-status handling for machine-shop operations.
- Supported completed, semi-finished, defect and plan-change workflows; integrated QR with FMS for process-level traceability.

Machine Shop IoT 4.0 & FMS Integration

- Contributed to IoT–FMS integration covering operator login, process selection, timer activities and machine operations.
- Worked on machine-based HMI and mobile workflows for production data capture and digital monitoring.

MPI Display, Inspection & Stage-Wise Photo Upload

- Developed and maintained the MPI Display Application (React.js, ASP.NET Web API) for inspection visibility.
- Built the Stage-Wise Product Image module with **Pre-Machining** and **Final Machining Weight Photo** uploads, linking images to product and process references.
- Reduced dependency on paper-based records by enabling digital evidence collection and retrieval.

AI Assistant – PO Register Automation

- Developed an AI-assisted workflow with PDF upload, document reading and purchase-order field extraction.

- Implemented validation, existing-PO checks, customer identification, Work Order number generation and PO PDF archival.
- Reduced repetitive manual PO registration through end-to-end automation across frontend, APIs and SQL Server.

Man-Hours-per-Ton Dashboard & Power BI Displays

- Extended the dashboard to the SS Area (SS-Rough, SS-Visual, SS-DP, SS-Final); improved Welding and Grinding analysis and booth-wise reporting.
- Worked on SQL query logic, API integration and FP-number-wise comparison reporting.
- Built a web-based Power BI display page that auto-rotates report views on shared production screens.

DP Requirements System & Application Support

- Developed the DP Requirements web application (React.js, ASP.NET Web API) for digital requirement management.
- Support application publishing, deployment troubleshooting, system administration and user requirement gathering.

Junior Software Developer — July 2022 – December 2023

Longwood Infotech, Coimbatore

- Developed web application features using C#, ASP.NET and Microsoft SQL Server, and frontend components with React.js.
- Built a React.js e-commerce website with Material UI for product presentation and sales workflows.
- Wrote and modified SQL queries; integrated frontend features with application logic and databases.
- Participated in requirement analysis, development, debugging, testing and deployment across the SDLC.

KEY PROJECTS

AI Assistant – PO Register Automation

Technologies: React.js, ASP.NET Core Web API, C#, SQL Server, PDF Processing

- Automated purchase-order PDF processing: field extraction, validation, Work Order generation, PO Register insertion and archival.

FMS Process Entry Web Application

Technologies: React.js, ASP.NET Core Web API, C#, SQL Server, Material UI

- Responsive shop-floor data capture with centralized recording and retrieval of manufacturing process information.

QR Scanner & Shop-Floor Traceability

Technologies: React.js, ASP.NET Web API, SQL Server

- QR-based scanning and production status updates giving process-level traceability through FMS integration.

Machine Shop IoT 4.0 Integration

Technologies: ASP.NET Web API, SQL Server, IoT–FMS Integration

- Digital workflows for login, process selection, timers and production data capture via HMI and mobile devices.

Stage-Wise Product Image & Weight Photo Management

Technologies: React.js, ASP.NET Web API, SQL Server, File Upload

- Digital evidence management for casting, machining and inspection stages with weight-photo uploads.

Man-Hours-per-Ton Dashboard

Technologies: React.js, ASP.NET Web API, SQL Server, Reporting Integration

- Manufacturing effort analytics covering SS Area, Welding, Grinding and booth-wise performance.

MPI Display, DP Requirements, Power BI Auto-Rotating Dashboard

Technologies: React.js, ASP.NET Web API, C#, SQL Server, HTML/CSS/JS, Power BI

- Inspection display, requirement management and timed report rotation for shared display screens.

React.js E-Commerce Website & Personal Portfolio

Technologies: React.js, JavaScript, Material UI, HTML5, CSS3

- Product-oriented e-commerce UI with reusable components; portfolio site showcasing skills and projects.

EDUCATION

Bachelor of Computer Science — June 2019 – April 2022

Hindusthan College of Arts and Science, Coimbatore

INTERNSHIP & CERTIFICATION

Java Intern — Oct 2021 – Jan 2022

Net Tel Solutions India Pvt Ltd (Excellence in Services)

Full Stack Developer Online Course (Java) — Jan 2023 – Oct 2023

Besant Technologies, Chennai

Languages: English, Tamil